



Parul University
Faculty of Engineering and Technology
Department of Applied Science & Humanities
Academic Year 2024-25
Subject: Quant and Reasoning (303105311)
Branch: CSE/ IT

UNIT-5

QUESTION BANK

1. 43 men can do a work in 89 days. Find the amount of work done (in %) by 90 men in 55 days.
2. 10 men can do a work in 12 days. Find the time required to do 20% of the work working twice as fast as before.
3. Twenty women can do a work in sixteen days. Sixteen men can complete the same work in fifteen days. What is the ratio between the capacity of a man and a woman?
4. 10 women can complete a work in 7 days and 10 children take 14 days to complete the work. How many days will 5 women and 10 children take to complete the work?
5. A is twice as fast as B. If B alone can do a piece of work in 30 days, in what time can A and B together complete the work?
6. A pipe can empty a cistern in 40 minutes. Find the time in which $\frac{3}{4}$ part of the cistern will be emptied.
7. A tap can fill a cistern in 8 hours and another can empty it in 16 hours. If both the taps are opened simultaneously, find the time (in hours) to fill the cistern.
8. One fill pipe A is 10 times faster than the second fill pipe B. If B can fill a cistern in 55 minutes, then find the time when the cistern will be full if both fill pipes are opened together.
9. A pipe can fill a tank in 12 hours. Due to leakage at the bottom, it is filled in 24 hours. If the tank is full, how much time will the leak take to empty it?
10. A tap can fill a tank in 25 minutes and another can empty it in 50 minutes. Find out whether the tank will be filled up or emptied in how many minutes?
11. Two pipes can fill a tank in 10 hours and 12 hours, respectively. While a third pipe emptied the full tank in 20 hours. If all the three pipes operate simultaneously, in how much time the tank will be filled?
12. Two pipes can fill a cistern separately in 24 minutes and 40 minutes respectively. A waste pipe can drain off 30 litres per minute. If all the three pipes are opened, the cistern fills in one hour. Find the capacity (in litres) of the cistern.
13. While going to office, Ramesh travels at a speed of 30 kmph and on his way back, he travels at a speed of 45 kmph. What is his average speed of the whole journey?
14. A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?
15. A man complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.
16. The ratio between the speeds of two trains is 7:8. If the second train runs 400 km in 4 hours, then find the speed of the first train.
17. Sound travels 330 metres a second. If the sound of a thunder cloud follows the flash after 10 seconds, Find the thunder cloud is at a distance.



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18. A train travels 92.4 Km/h. How many metres will it travel in 10 minutes?
19. A person covers half of his journey at 30 Km/h and the remaining half at 20 Km/h. The average speed for the whole journey is ?
20. Rajesh covers a certain distance by bus at 16 Km/h and returns at the starting point on a cycle at 9 Km/h. His average speed for the whole journey is?
21. A train passes a 150 m long railway bridge in 18 seconds. If the train is running at a speed of 60 Km/h., then find the length of the train in metres.
22. A 100 metres long train passes a bridge at the rate of 72 Km/h in 25 seconds. What is the length of the bridge?
23. A train covers a distance in 50 minutes if it runs at a speed of 48 Km/h. The speed at which the train must run to reduce the time of journey to 40 minutes, will be?
24. Rakesh sets out to cycle from Delhi to Mathura and at the same time Suresh starts from Mathura to Delhi. After passing each other, they complete their journeys in 9 and 16 hours, respectively. At what speed does Suresh cycle if Rakesh cycles at 16 Km per hour?
25. A train is running at a uniform speed of 60 Km/h. It passes a railway platform in 15 seconds. If the length of the platform is 130 m, then the length of the train is?
26. A boat goes 13 Km upstream in 39 minutes. The speed of stream is 3 Km/h. Find the speed of boat in still water.
27. A swimmer covers a distance of 28 Km against the current and 40 Km in the direction of the current. If in each case he takes 4 hours, then find the speed of the current.
28. A boat travels 2 Km upstream in a stream flowing at 3 Km/h and, then returns downstream to the starting point in 30 minutes. Find speed of the boat in still water.
29. A boat can travel with a speed of 13 km/hr in still water. If the speed of stream is 4 km/hr in the same direction, then find time taken by boat to go 63 km in opposite direction.
30. A motor-boat, travelling at the same speed, can cover 25 Km upstream and 39 Km downstream in 8 hours. At the same speed, it can travel 35 Km upstream and 52 Km downstream in 11 hours. Find the speed of the stream.